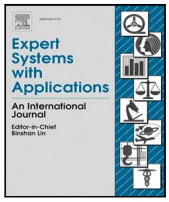
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SGFNet: A semantic graph-based multimodal network for financial invoice information extraction

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ABSTRACT

To meet the demand for a large amount of invoice entry work in the financial industry and improve the low accuracy of traditional manual entry, we construct SGFNet, a financial invoice information extraction network that integrates semantic graph associations and multimodal modeling. First, we construct a graph of strong and weak semantic associations between data within each modality based on the correlation of text content. Subsequently, we model the multimodal data in a unified structure, extract the text modal information of invoices along with corresponding image and layout modal information, and guide the fusion and embedding of multimodal data through semantic associations in the graph to produce a richer feature representation. Furthermore, semantically linked multimodal information is fed into an aggregated multimodal self-attention mechanism to establish effective connection between modalities. Finally, with the combination of supervised contrastive learning and smoothed Kullback–Leibler divergence in terms of loss functions, the accuracy degradation problem incurred by sample imbalance and convergence instability is reduced. In our experiments, we achieved F1 scores of 93.71% for the English financial invoice dataset and 96.27% for the Chinese dataset, indicating that the proposed method successfully extracts feature information from different data modalities, thereby achieving satisfactory information extraction results.

1. Introduction

Financial invoices have become an essential component of modern business transactions owing to a continuous increase in commercial activities. As commercial documents, financial invoices record important information about commercial activities, including the prices, product quantities, dates, and parties involved in the transaction of goods and services. Invoices play a crucial role in the financial field by not only proving the authenticity of business activities, but also providing enterprises with the basis for financial reports, tax returns, and other important tasks. In certain countries and regions, invoices serve as a basis for tax payments (Park, Cho, & Lee, 2023; Zhang, Zheng, Dong, & Feng, 2021). It is therefore crucial to accurately and efficiently extract key information from financial invoices.

However, extracting information from semi-structured invoices in the financial sector remains highly challenging due to the heterogeneity of said invoices in their formats and contents. Moreover, most invoices are reviewed and filtered manually, which is inefficient given their large volume. The conventional approach also presents obstacles to the systematic management of the business market and digital development. Therefore, the efficient and accurate extraction of key information from semi-structured documents has become a popular research topic.

Several recent studies have focused on extracting information from receipts and invoices by capturing key information in the data. However, these studies did not target specific financial invoices as research subjects. Financial invoices differ from receipts in terms of content because they are legal documents used to prove the legitimacy of commercial transactions. In contrast, a receipt is generally used only to prove the completion of a transaction and payment. Consequently, invoices are more valuable for research than receipts in the financial sector. Additionally, annotated benchmark financial invoice datasets are typically unavailable owing to the confidentiality of information. Although receipts and invoices share some common properties, their analysis differs significantly owing to the complex graphical layout and richer content of information represented in the latter. A recent study analyzed financial reports and confidentiality agreements, including charitable organization documents, and found that state-of-the-art models exhibit significant performance degradation when oriented toward documents with complex layouts (Stanisławek et al., 2021).

To address these challenges, we propose SGFNet, a network model that fuses semantic and multimodal information to extract information

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from semi-structured financial invoices. The model differs from previous studies by targeting financial invoices with significant research value and combines semantic information from modalities in the multimodal information fusion phase to better capture key information from financial invoices with complex layouts. Here, we present each module and its improvements in detail. First, we model the multimodal data in a unified structure, separately extract information of different modalities, and construct links between semantic linkage graphs within the multimodal information. These links are used in the interaction between information modalities, thereby improving the intra-modal and intermodal information associations. The semantically linked multimodal information is then fed into a newly constructed multimodal self-attention mechanism, and the multimodal features are fused to establish effective links between modalities. Finally, the accuracy and generalizability of financial invoice entity extraction are optimized by combining supervised comparative learning and a cross-entropy loss function, minimizing the accuracy degradation stemming from sample imbalance. The proposed SGFNet network model is well-suited for extracting information from Chinese financial invoices. The remained of the paper is organized as follows. Section 2 outlines the current research status of invoice information extraction. Section 3 describes the SGFNet network model and its various modules in detail. Section 4 presents the experimental results and analysis, and Section 5 concludes the paper.

2. Related work

A range of techniques is utilized to support and expedite the crucial process of extracting information from different invoices (Ha & Horák, 2022). Current smart invoice information extraction methods can be classified as rule- and template-based methods, statistical machine learning-based methods, and deep learning-based methods.

Traditional approaches employ rules and templates to extract information from invoices (Cesarini et al., 1997). Cesarini, Francesconi, Gori, and Soda (2003) classified rules into different types of knowledge according to vendor dependencies, and proposed a multi-label invoice analysis system that connects layout information to textual information. Klein, Dengel, and Fordan (2004) proposed a method called SmartFix, which uses case-based inference techniques to construct layout graphs (Hamza, Belaïd, & Belaïd, 2007), thereby distinguishing between known and unknown classes of invoices. However, the limited expert ruleset is ineffective when comprehensively and systematically analyzing a large variety of knowledge data. As a template-based approach, Bart and Sarkar (2010) proposed a template-matching method that obtains key information from invoices by matching the user-entered form or entry with the content and structure in a given template in terms of height, width, size, and other features. By considering time and resource costs, Esser, Schuster, Muthmann, and Schill (2014) designed an invoicing system for small businesses and individuals by grouping documents of the same category and generating different matching rules for each group based on the number of samples within that group, thereby obtaining indexed combinations with different values and positions. By conducting experiments, they achieved an F1 score of 78% for information extraction.

The statistical machine learning-based approach treats the invoice information extraction task as a sequence labeling problem, extending the application of information extraction to large-scale data. Aslan, Karakaya, Unver, and Akgul (2015) used a two-stage optimization structure to solve the invoice information processing and information extraction problem, first using a support vector machine (SVM) and histogram of oriented gradient (HOG) – alongside other methods – to generate candidate objects for different types of invoices, and then parsing the invoices in a combinatorial arrangement. Hong, Yeo, Cho, and Ahn (2018) used a random forest (RF) to automatically identify core suppliers from electronic invoice data in a key supplier information extraction task, improving the efficiency of supplier segmentation. Tang et al. (2020) used methods such as k-means and the multilayer perceptron (MLP) to automatically determine anomalous behavior, with potential threats correlated to identify anomalies occurring in the electronic invoicing system, thereby mining key information with threats in the invoices.

Efficient deep learning methods are increasingly being utilized to extract invoice information due to continuous advancements in deep learning technology. Palm, Winther, and Laws (2017) proposed an invoice system combining recurrent neural networks with long- and short-term memory networks to learn item dependencies through continuous feedback from users, obtaining satisfactory information extraction metrics. Subsequently, Palm, Laws, and Winther (2019) proposed an invoice recognition system called InvoiceNet, which was based upon a special deep neural network architecture, and performed information extraction tasks by designing extended convolutional block sets. The experimental results demonstrated improved information extraction performance for four types of invoices. Tarawneh, Hassanat, Chetverikov, Lendak, and Verma (2019) proposed a method to extract features from invoice images by combining the convolutional neural network (CNN) Alexnet with statistical machine learning methods, with invoices initially classified into three categories according to their types before recognition. Their experimental results showed that the k-nearest neighbors (KNN) algorithm outperforms the RF and Naive Bayesian (NB) methods. JADLI and Mustapha (2020) tested various types of pre-trained CNN models on a dataset containing both invoices and receipts for the receipt and invoice information extraction problem (DenseNet, etc. Huang, Liu, Van Der Maaten, & Weinberger, 2017) with classification algorithms combining different dimensionality reduction methods. They achieved a maximal accuracy of 96.1% using a combination of VGGNet (Simonyan & Zisserman, 2015) and logistic regression.

With the increasing prevalence of graph neural networks, some studies have implemented these networks for invoice recognition (Kipf & Welling, 2016; Yu, Lu, Qi, Gong, & Xiao, 2021). Riba et al. (2019) employed the underlying concept of graph neural networks to construct a form layout detection method for invoice documents without using raw text – instead using layout, context, and category information to represent the structure - and achieved good results on two invoice datasets. Lohani, Belaïd, and Belaïd (2019) used graph convolutional networks to implement a system that processes digitized invoice images, decomposes invoice data into different entity types, and then extracts entity information through graph structures. Liu, Gao, Zhang and Zhao (2019) used a graph convolutional architecture to extract invoice information by first training graph embeddings to learn textual contextual information, and then introducing textual embeddings to extract entity information, outperforming the BiLSTM-CRF baseline model. Krieger, Drews, Funk, and Wobbe (2021) used a graph convolutional network that accounts for layout information to extract fragments of interest from invoices, achieving an F1 score of 87.53%.

In addition to other approaches, multimodal deep learning methods have also been utilized in the field of invoice information extraction (Huang, Lv, Cui, Lu, & Wei, 2022; Xu et al., 2020, 2021). In one study, textual information was extracted from invoices using a CNN that transforms and embeds semantics in text into a class-lattice structure while considering positional modal features (Zhao, Niu, Wu, & Wang, 2019). Another study proposed an approach that integrates multimodal information features with a large-scale pre-trained model, enabling the model to perform tasks such as document image classification and receipt interpretation. However, this study did not consider the underlying semantic relationships between text segments. Utilizing the concept of the transformer network structure from BERT (Appalaraju, Jasani, Kota, Xie, & Manmatha, 2021; Kenton & Toutanova, 2019) developed similar models trained on large quantities of data before being fine-tuned for use in downstream tasks.

In summary, although many studies have applied deep learning to invoice information tasks, few of them specifically focused on information extraction of fine-grained financial invoices or semantic

associations multimodal data. Because these models fail to sufficiently master local key feature information and are prone to information loss, they exhibit a suboptimal invoice information extraction accuracy. In this study, we employ the multimodal idea of semantic association in the context of financial invoice information extraction to generate richer feature representations and construct a multimodal network incorporating semantic graphs, thereby improving the effectiveness of invoice information extraction.

3. Methodology of the proposed solution

We present SGFNet as a multimodal neural network based on a semantic graph designed to extract financial invoice information. First, we construct a semantic graph to establish associations between different data within the same modality based on the strength of their textual content correlations. Then the information of different modal data is modeled in a unified structured way. Invoice data feature maps are extracted as image modal embedding features using a visual coder, and text modal information and layout modal information are extracted using an optical character recognition (OCR) system. We use the Euclidean distance between the center coordinates as the layout information feature and the textual subtext as the text feature. Furthermore, we calculate the Hadamard product of different data within the text and image modalities and multiply it by the defined association strength coefficients as the token, and then establish the association between the data within the modalities for the layout modal information. In addition, the input data are utilized to enhance the interactions between modalities using a novel multimodal self-attention layer to establish effective correlations between modalities. Finally, the financial invoice information extraction results are optimized by iterating on the classifier using an improved loss function. An overall flowchart of the proposed SGFNet model is presented in Fig. 1.

3.1. Semantic graph construction

Because financial invoices often contain multiple types of interrelated textual content, we propose an approach that constructs a semantic graph of the invoice information entities. Based upon associations between textual content, this graph can be used to capture hidden correlations between data within text and image modalities. This semantic graph is undirected, with nodes representing invoice information entities and edges representing semantic relationships between the entities. For textual data, we pre-define the corresponding entities; classify the entity categories into "title", "question", "answer" and "other;" annotate the entities, and represent them as nodes in the semantic graph. We then establish connections between entities with semantic relationships, as expressed by Eq. (1). We define e_{ij} as the strength of the semantic relationship between two entities, where $i = 1, 2, 3, \ldots, n$, $j = 1, 2, 3, \ldots, m$. A e_{ij} value of 1 indicates a strong semantic association between two entities, whereas a value of 0.5 indicates a weak semantic association, and a value of 0 indicates no association. An example of a constructed semantic graph is shown in Fig. 2, depicting an invoice containing n questions and m answers, where Q_i and A_j denote the question and answer entity categories, respectively. We add edges according to strong and weak semantic associations between entities, denoted by strong and dashed lines, respectively. For example, because "Gas mask" and "Gate valve" both belong to "Name of goods or taxable labor and services", the association between them is considered strong. Although an inclusion relationship exists between the abstract entity "Name of goods or taxable labor and services" and concrete trade names, the two do not belong to the same entity type, and the semantic association between them is considered weak.

$$e_{ij} = \begin{cases} 1 & \text{When the strong semantic association} \\ 0.5 & \text{When the weak semantic association} \\ 0 & \text{When no semantic association} \end{cases} \quad (1)$$

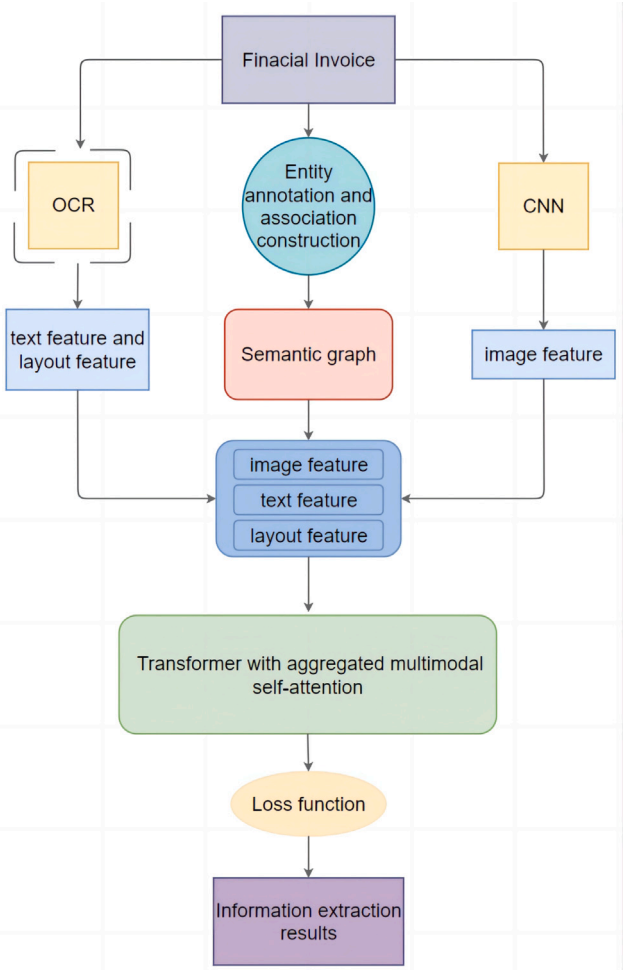


Fig. 1. Flowchart of SGFNet model.

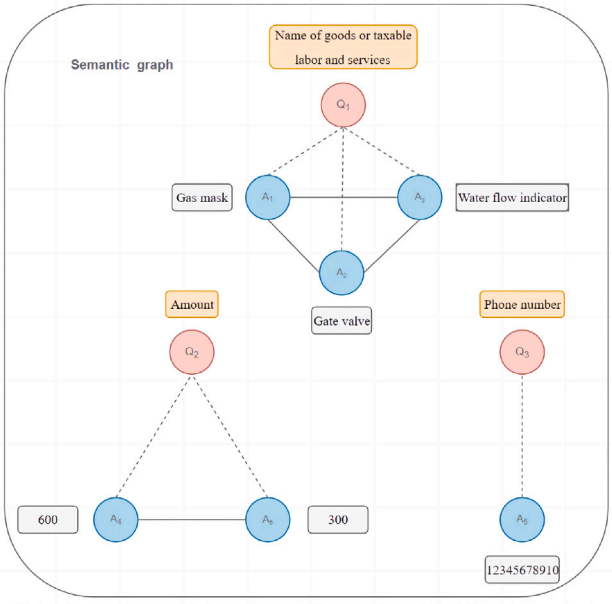


Fig. 2. Example of semantic graph.

3.2. Multimodal feature fusion modeling

Deep learning-based information extraction methods designed for financial invoices generally employ data preprocessing methods to first extract information of various modalities, then convert the input information into a distributed representation using specific encoding methods, and finally input the representation into an encoder for training to obtain results. However, existing methods do not combine feature extraction with semantic information, resulting in relatively shallow feature representation. We therefore propose a multimodal feature fusion modeling approach guided by semantic graphs, wherein each modal feature is initially extracted separately, and the features are subsequently associated and aligned through semantic graphs to improve semantic associations within modalities while enhancing the interaction between modalities and enriching the feature representation. In the following subsections, we describe how to perform feature extraction and fusion in text, image, and layout modalities.

3.2.1. Textual feature extraction

We employ the PP-OCRv3 (Li et al., 2022) annotation tool provided by the Baidu Paddle platform to extract textual data from financial invoices, convert textual information into tokens, perform Hadamard product operations on associated tokens through the semantic graph, and align tokens according to semantic information. Because Chinese words are not delineated by spaces, the textual data are tokenized using SentencePiece (Kudo & Richardson, 2018) with a unigram language model to unify the text input of multiple languages. Thus, we obtain the tokenized vector $t_{\text{tok}} = \left\{[CLS], t_{Q_1} \odot t_{A_1} e_{11}, t_{Q_1} \odot t_{A_2} e_{12}, \ldots, t_{Q_n} \odot t_{A_m} e_{nm}, \ldots, t_{512}\right\}$, where $\odot$ denotes the Hadamard product operation and $[CLS]$ denotes the common starter used to classify the text. If the length of the sequence of t_{tok} is less than 512, it is complemented by the $[PAD]$. Then t_{tok} is passed through a trainable embedding matrix W_T to obtain the text embedding $\bar{T}$, as shown in Eq. (2):

$$T = W_T \left\{[CLS], t_{Q_1} \odot t_{A_1} e_{11}, \ldots, t_{Q_n} \odot t_{A_m} e_{nm}, \ldots, t_{512}\right\} \tag{2}$$

3.2.2. Image feature extraction

Let the two arbitrary rectangular character image blocks in the sample be $V_a, V_b \in R^{3\times h\times w}$, where h is the pixel height of the rectangular image block, w is the pixel width of the image block, $a, b = 1, 2, \ldots, n'$, and n' is the number of rectangular character image blocks in any sample. V_a and V_b are fed into the ResNet50 (He, Zhang, Ren, & Sun, 2016) network and the feature map $V_{al4}, V_{bl4} \in R^{c\times h_{l4}\times w_{l4}}$ is extracted from its layer 4, where c is the number of channels, and h_{l4} and w_{l4} are the height and width of the features in that layer. In this paper, this extraction operation is represented by the functions $f_{resnet50}(V_a)$ and $f_{resnet50}(V_b)$. Since the input of Transformer encoding layer is a linear vector, this paper uses a 1×1 convolution kernel with channel number $d = 768$ to convolve V_{al4} and V_{bl4}, and flattens them into a one-dimensional vector with pixel length $N = 512$, and finally calculates the Hadamard product between them. The image feature embedding is shown in Eq. (3), where FL denotes the matrix flattening operation and $con_{d\times1\times1}$ denotes the convolution operation:

$$\begin{aligned} V = {} & FL(con_{d\times1\times1}(f_{resnet50}(V_a)))\odot \\ & FL(con_{d\times1\times1}(f_{resnet50}(V_b))) \cdot e_{ab} \end{aligned} \tag{3}$$

3.2.3. Layout feature extraction

For the layout modal information, we use OCR to extract it as well as the text modal. For two arbitrary rectangular image blocks identified in a sample, this paper extracts their center coordinates information, and then calculates the cosine distance information between them as the layout modal feature embedding in the encoding stage. Since the OCR system obtains the coordinates of the four corners of a rectangular character image block, represented as $\{(x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4)\}$ and $\{(x_1', y_1'), (x_2', y_2'), (x_3', y_3'), (x_4', y_4')\}$ in a set, its center coordinates can be obtained as $((x_1 + x_3)/2, (y_1 + y_3)/2)$ and $((x_1' + x_3')/2, (y_1' + y_3')/2)$. Therefore, the distance $\overline{D}$ between the center points of two image blocks can be defined as follows:

$$\overline{D} = \frac{(x_1 + x_3)^2 + (x_1' + x_3')^2 - (y_1 + y_3)^2 - (y_1' + y_3')^2}{2} \tag{4}$$

As a result, we can obtain the layout features of each image block extracted from the input document, which is denoted by L in Fig. 3. We can then obtain the layout modal embedding of the image and text, respectively, as shown in Eqs. (5) and (6):

$$\overline{V}_L = W_V^L \overline{D} + P_V^{abs} \tag{5}$$

$$\overline{T}_L = W_T^L \overline{D} + P_T^{abs} \tag{6}$$

where W_V^L denotes the trainable embedding matrix for image encoding, W_T^L denotes the trainable embedding matrix for text encoding, P_V^{abs} and P_T^{abs} denote the absolute 1D position encoding commonly used on images and texts, and $\overline{V}_L$ and $\overline{T}_L$ denote the image and text layout feature embedding. The absolute position encoding is combined with various layout features in the embedding process to capture more fine-grained layout feature information. The semantic graph guides the fusion of multimodal information and then aligns them with each other to achieve a unified and structured modeling of multimodal features, resulting in a richer feature representation and thus improving the feature extraction capability of the network.

3.3. Aggregated multimodal self-attention

Many studies have demonstrated that the self-attention mechanism can achieve satisfactory results when extracting features of different modalities. However, the self-attention mechanism can only implicitly capture the relationship between tokens and location information, which is not conducive to the association of multimodal information. To efficiently fuse multimodal information and enhance interactivity between different modal features, we employ an aggregated multimodal self-attention, which computes the aggregation of encoded embedded multimodal features to establish effective associations between modalities.

The self-attention mechanism is expressed in Eq. (7), where the query matrix encoded by W^Q and W^K is multiplied by the transpose matrix of key and normalized, and then multiplied by the value encoded by W^V to obtain the feature matrix. Let $i, j \in 1, 2, \ldots, N$, multiply the input by Eq. respectively to obtain the self-attention feature output S_i, d denotes the dimension of query and key:

$$S_i = \sum_{j=1}^{N} \frac{\exp(\frac{1}{\sqrt{d}}(W^Q x_i) \cdot (W^K x_j)^T)}{\sum_{j'=1}^{n} \exp(\frac{1}{\sqrt{d}}(W^Q x_i) \cdot (W^K x_{j'})^T)} (W^V x_j) \tag{7}$$

From Eq. (7), we can see that it extracts the features of a single modality itself, but does not establish the connection between the information of each modality, so this paper introduces the layout feature information in image and text embedding from the perspective of key and query to enhance the interaction between features. Let $i = 1$, then we can derive that:

$$\delta^v = (W_v^Q x_i^v)(W_v^K x_j^v)^T + (W_L^Q \overline{V_L})(W_L^K \overline{V_L})^T \tag{8}$$

$$\delta^t = (W_t^Q x_i^t)(W_t^K x_j^t)^T + (W_L^Q \overline{T_L})(W_L^K \overline{T_L})^T \tag{9}$$

where x_i^v, x_j^v, x_i^t, x_j^t denote the embedding features of image and text, W_v^Q, W_v^K denote the learnable matrix of image embedding of query and key, W_t^Q, W_t^K denote the learnable matrix of text embedding of query and key, W_L^Q, W_L^K denote the learnable matrix of query, key layout modal embedding. Thus we can obtain the feature output M_i of

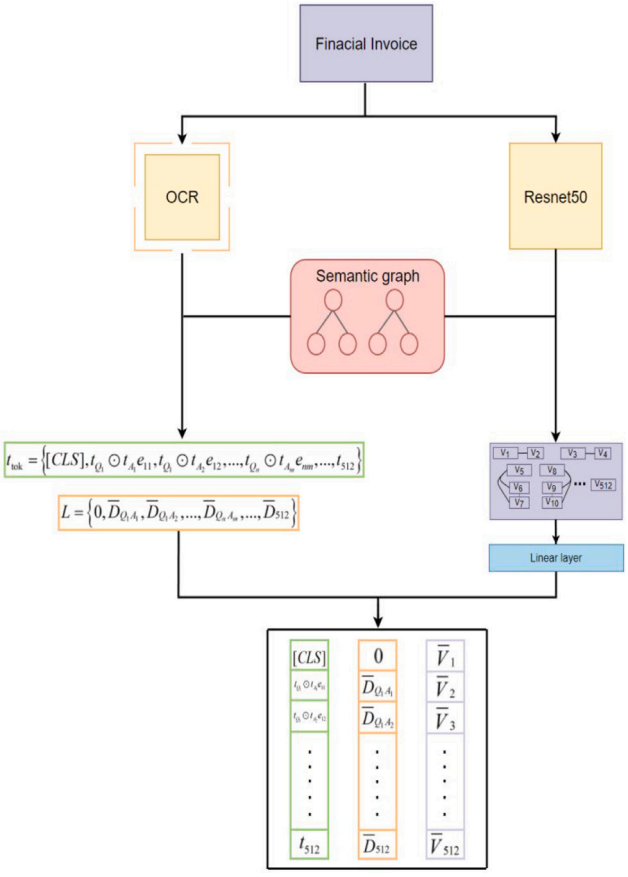


Fig. 3. Flowchart of multimodal feature fusion modeling.

the multimodal self-attention mechanism in this paper, as shown in the following equation:

$$M_i = \sum_{j=1}^{N} \frac{\exp(\frac{1}{\sqrt{d}}\delta^v)}{\sum_{j'=1}^{n}\exp(\frac{1}{\sqrt{d}}\delta^v)}(W^V x_j^v) + \sum_{j=1}^{N} \frac{\exp(\frac{1}{\sqrt{d}}\delta^t)}{\sum_{j'=1}^{n}\exp(\frac{1}{\sqrt{d}}\delta^t)}(W^V x_j^t) \tag{10}$$

Comparing Eqs. (7) and (10), we can see that the proposed aggregated multimodal self-attention can aggregate the features of layout, text, and image modal information simultaneously compared with the initial self-attention mechanism so that the network can focus on richer information and reduce the information loss, and share the weights between image and text modalities in terms of input features, which enhances the feature fusion ability of the network and improves the correlation between different modalities.

3.4. Information classification

The feature output of the attention module is computed for classification to extract invoice entity information. For the information extraction task, the cross-entropy loss function is generally used to calculate the difference between true and predicted values, obtaining a result that meets the expectation through continuous iterations. However, large gaps may exist between certain entity types in financial invoice data, leading to a data imbalance. For example, data labeled "question" and "answer" generally occur much more frequently than those labeled "title". The cross-entropy loss function often exhibits unstable convergence when encountering such data (Gunel, Du, Conneau, & Stoyanov, 2020). To address this problem, we combined supervised contrast learning with the concept of the smoothed Kullback–Leibler divergence in terms of the loss function to improve model stability during convergence, thereby achieving information classification.

Consider the category labels of the entities in the set $[1, 2, 3, \ldots, H]$, where H is the number of category labels. Letting P be the number of samples input in each training batch, and a_i and b_i denote the samples, $i = 1, \ldots, P$, we can obtain the cross-entropy loss function L_{CE} as follows:

$$L_{CE} = -\frac{1}{P}\sum_{i=1}^{P}\sum_{h=1}^{H} b_{i,h} \cdot \log\hat{b}_{i,h} \tag{11}$$

where $b_{i,h}$ denotes the category label (using ONE HOT representation), and $\hat{b}_{i,h}$ denotes the probability output of the ith word element belonging to class h. In this paper, we add the constraint of imbalanced data samples to the loss function, which distances the samples with different entity classes and shortens the distance between samples with the same entity class, and the supervised comparison learning loss function is obtained as follows:

$$L_{SCL} = \sum_{i=1}^{P} -\frac{1}{P_{b_i}}\sum_{j=1}^{P} f\, g \log\frac{\exp(\phi(a_i)\cdot\phi(a_j)^T/\tau)}{\sum_{p=1}^{P} l\cdot\exp(\phi(a_i)\cdot\phi(a_p)^T/\tau)} \tag{12}$$

where P_{b_i} denotes the number of samples with the same label as b_i for each batch of input samples, $\phi(\cdot)$ denotes the L2 normalization function, $\tau > 0$ is the adjustable hyperparameter, and f, g, and l denote the indicator functions of 1 when $i \neq j$, $b_i = b_j$, and $i \neq p$, respectively, and 0 otherwise. The loss function based on supervised comparison learning captures the similarity between similar samples in the invoice data and compares them with samples in other classes, so that the same samples in each batch of input samples can obtain lower loss and samples in different classes obtain higher loss values, thus better-distinguishing samples in different classes. To further avoid the effect of the cross-entropy loss function, we additionally add the smoothed KL divergence loss function to the supervised contrast learning loss function to improve the model convergence ability (Zhao, Li, Zhen, Huo, & Zhang, 2020). Define $Q(x)$ to denote the true label probability distribution and $P(x)$ to denote the predicted probability distribution, and both $Q(x)$ and $P(x)$ are Gaussian distribution functions. The closer the true category probability distribution and the predicted category probability distribution are, the better, so the KL divergence loss function is defined as shown in the following equation:

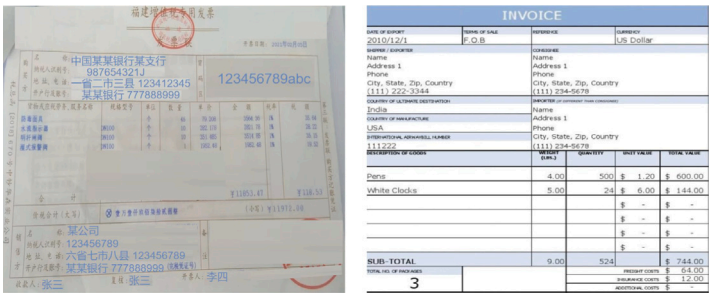
$$\begin{aligned} L_{KL} &= \int Q(x)(\ln\frac{Q(x)}{P(x)})dx \\ &= \int Q(x)\ln Q(x)dx - \int Q(x)\ln P(x)dx \end{aligned} \tag{13}$$

The following equation is derived from the properties of the Gaussian distribution function:

$$\begin{aligned} L_{KL} &= \int Q(x)\ln Q(x)dx \\ &\quad - \int Q(x)(-\ln(\sqrt{2\pi}\sigma) + \ln e^{\frac{-(x-x_e)^2}{2\sigma^2}})dx \\ &= \int Q(x)\ln Q(x)dx \\ &\quad + \int Q(x)(\frac{1}{2}\ln(2\pi\sigma^2) + \frac{(x-x_e)^2}{2\sigma^2})dx \end{aligned} \tag{14}$$

The Dirac function is defined as shown in Eq. (15), and since the Gaussian distribution is an approximation of the Dirac function when the standard deviation is close to 0, Eq. (14) is derived from the screening property of the Dirac function in Eq. (16) to obtain Eq. (17).

$$\int \delta(x)dx = 1 \tag{15}$$



(a) (b)

Fig. 4. Example of a financial invoice dataset. (a) Chinese Invoice; (b) English Invoice.

Table 1
Results of different models on the Chinese invoice dataset.

Model	Parameters (M)	Precision (%)	Recall (%)	F1 score (%)
SGFNet	180	95.06	97.51	96.27
StrucTexTv2	238	95.31	96.34	95.82
DocFormer	183	91.57	93.18	92.37
BROS	110	87.52	89.54	88.52
LayoutXLM	200	87.22	89.74	88.46
LayoutLM	160	78.64	80.36	79.49
RoBERTa	125	64.74	69.7	67.13

$$\int \delta(x - x_g)y(x)dx = y(x_g) \tag{16}$$

$$L_{KL} = \frac{\ln(2\pi)}{2} + \frac{\ln(\sigma^2)}{2} + \frac{(x_g - x_e)^2}{2\sigma^2} - H(Q(x)) \tag{17}$$

Since the constant terms have no effect on the derivative, the terms without parameters can be rounded off to obtain the following equation:

$$L_{KL} \propto (\frac{\ln(\sigma^2)}{2} + \frac{(x_g - x_e)^2}{2\sigma^2}) \tag{18}$$

If the initial value of σ in Eq. (18) is large, it will easily lead to gradient explosion in the initial stage of training and thus the model cannot converge normally, and there is the problem of input limitation in the mathematical calculation of ln, therefore, in this paper, we make the variables α and σ satisfy the relationship Eq. (20) in the prediction stage of model training, and then bring Eq. (19) into Eq. (18) to obtain the loss function as shown in Eq. (20).

$$\alpha^2 = \ln \sigma^2 \tag{19}$$

$$L_{KL} \propto (\frac{\alpha^2}{2} + \frac{e^{-\alpha^2}(x_g - x_e)^2}{2}) \tag{20}$$

To further enhance the robustness of the model, the KL loss function is smoothed in this paper, and the KL divergence loss function of the model is obtained when $\left|x_g - x_e\right| > 1$:

$$L_{KL} = \mathrm{e}^{-\alpha^2}\left|x_g - x_e\right| + \frac{\alpha^2}{2} \tag{21}$$

Ultimately, in this paper, the supervised contrast learning loss function is combined with the KL divergence loss function as the loss function for the category:

$$L_{\text{total}} = L_{SCL} + L_{KL} \tag{22}$$

Algorithm 1 Procedure of the SGFNet model

```
function SGFNet(Documents):
// Extract Features
    for Doc in Documents:
        ImgFeatVec = ExtractImageFeatures(Doc.image)
        Append(ImgFeatVec, ImageFeatureVectors)
        Tokens = Tokenize(Doc.text)
        TextTokenSequences.append(Tokens)
        LayoutFeatVec = ExtractLayoutFeatures(Doc.
            layout)
        Append(LayoutFeatVec, LayoutFeatureVectors)
// Construct Semantic Graph
    for i in range(len(ImageFeatureVectors)):
        Node_i = CreateNode(ImageFeatureVectors[i])
        AddNode(SemanticGraph, Node_i)
// Connect nodes based on feature similarity and layout
     information
        for j in range(i+1, len(ImageFeatureVectors)):
            Edge_ij = CreateEdge(Node_i, Node_j,
                Similarity(token[i], token[j]))
            if Edge_ij != NULL:
                AddEdge(SemanticGraph, Edge_ij)
// Embedding with Graph Guidance
    for i in range(len(Documents)):
        TextEmb = HadamardProduct(TextTokenSequences[i
            ], TextTokenSequences[j])*Similarity(token
            [i], token[j])
        ImgEmb = HadamardProduct(ImageFeatureVectors[i
            ], ImageFeatureVectors[j])*Similarity(
            token[i], token[j])
        LayEmb = ProjectLayoutFeature(
            LayoutFeatureVectors[i])
        CombinedEmb = CombineFeatures([ImgEmb, TextEmb,
            LayEmb])
        Append(CombinedEmb, MergedFeatureVectors)
// Aggregated Multimodal Self-Attention
    for Emb in MergedFeatureVectors:
        AttendedEmb = MultiModalSelfAttention(Emb)
        Append(AttendedEmb, output_features)
// Information Classification
    ClassLabels = ClassifyFeatures(output_features)
    return ClassLabels
```

The loss function based on the smoothed KL divergence makes the noisy samples not produce abrupt changes during the model training process, thus reducing their interference in the backpropagation process and making the model convergence more stable. The loss function with supervised contrastive learning and smoothed KL divergence reduces the accuracy degradation caused by sample imbalance while improving the stability of convergence during model training and further enhancing the accuracy of invoice information extraction. The pseudocode for the procedure of the SGFNet model is shown in Algorithm 1.

4. Experiments

4.1. Experimental environment

The deep learning platform in this paper is built on a server. The software environment of the model is Ubuntu 18.04 operating system and PyTorch 1.8.1 deep learning framework, and the programming language is Python 3.8. Hardware environment: CPU Xeon(R) Gold 6130, GPU NVIDIA Tesla V100-32 GB, memory size 25 GB.

4.2. Evaluating indicator

For the task of information extraction, this paper evaluates the performance of the financial invoice information extraction model using recall, accuracy, and $F1$ Score which takes the first two into account. The recall rate is sample-oriented and is used to describe the number of positives identified among all positive samples in the prediction process and is calculated as follows:

$$R = \frac{TP}{TP + FN} \times 100\% \tag{23}$$

where R is the recall rate, TP is the number of positive classes predicted as positive, and FN is the number of positive classes predicted as negative, which means the number of false identifications. The precision rate is oriented to the final predicted identification result and describes the ratio of predicted positive cases to all positive cases. The calculation formula is as follows:

$$P = \frac{TP}{TP + FP} \times 100\% \tag{24}$$

where P is the precision rate, FP is the number of samples in which negative cases are predicted to be positive cases. However, in most cases, it is a struggle to maintain a high level of recall and accuracy at the same time, so a metric is needed to reconcile these two parameters. In this paper, we use the score to measure the performance of the neural network model, which applies to multi-label information classification. The formula is as follows:

$$F1 = \frac{2 \cdot Recall \cdot Precision}{Recall + Precision} \times 100\% \tag{25}$$

When the conflict between recall and accuracy makes it impossible to judge the performance of the model, the score metric can better judge the performance of the model in the information extraction task.

4.3. Data preprocessing

The financial invoice dataset in this paper comes from a large commercial bank and the Internet. It encompasses 130 Chinese invoices and 50 English invoices, with examples shown in Fig. 4. All sensitive data have been desensitized. The original images are relatively large in size. If the original size is used as the training dataset, it will lead to excessive parameters. Therefore, the original invoice image size is reduced to 256 pixels × 256 pixels by pixel transformation. And based on the original images, the images are cropped and the brightness is changed so that the invoice images have different scale and presentation, which helps to avoid the occurrence of overfitting and thus improves the generalization ability of the training network. Fig. 5 shows the images in different forms after preprocessing.

Following preprocessing, 1000 financial invoice images are obtained, representing 700 Chinese and 300 English invoices. The images are allocated between training, testing, and validation sets according a 7:2:1 ratio. In this study, we imitate the format of the FUNSD (Jaume, Ekenel, & Thiran, 2019) dataset and fine-tune PP-OCRv3 to annotate the images automatically. Any labels with omissions are annotated manually. We use “title”, “question”, “answer”, and “other ” to label textual information, where “title” represent the beginning of each invoice (e.g., “INVOICE”), “question” represent the prefix of each part of the invoice to be completed (e.g., “DESCRIPTION OF GOODS”, “UNIT VALUE”), “answer” is used to mark each part of the invoice to be completed (e.g., address or telephone number), and “other” is used to indicate components with low information value.

Table 2
Results of different models on the English invoice dataset.

Model	Parameters (M)	Precision (%)	Recall (%)	F1 score (%)
SGFNet	180	93.37	94.05	93.71
StrucTexTv2	238	90.66	93.67	92.14
DocFormer	183	90.51	91.13	90.82
BROS	110	89.53	91.25	90.38
LayoutXLM	200	88.4	90.91	89.64
LayoutLM	160	80.35	81.32	80.83
RoBERTa	125	67.28	71.64	69.39

4.4. Results and analysis

To compare the effects of different loss functions on the training stability of the model, we first trained 200 epochs simultaneously in the same experimental environment and observed the curve changes of the loss values to analyze the training stability of various loss functions on the same data set, and the experimental results are shown in Fig. 6. It can be seen from the figure that the SGFNet model converges smoothly with different loss functions, and the loss function with both supervised contrast learning and smoothed KL divergence converges faster and with lower loss values, indicating that the model can obtain better training stability with the improved loss function.

The information extraction results obtained from the baseline model and SGFNet are then compared on the financial invoice dataset. The baseline models used in this paper include RoBERTa (Liu, Ott et al., 2019), LayoutLM (Xu et al., 2020), LayoutXLM (Xu et al., 2022), BROS (Hong et al., 2022), DocFormer (Appalaraju et al., 2021), and StrucTexTv2 (Yu et al., 2023). RoBERTa is a network model that improves the training strategy and parameterization of BERT. LayoutLM jointly models the interaction between text and layout information in document images, and visual information is merged into the model. DocFormer is based on a multimodal transformer architecture that interacts across modal features between different modalities. LayoutXLM is a multimodal pre-trained model for multilingual document understanding designed to cross the language barrier in document understanding. BROS is a pre-trained language model that encodes the relative position of text in a two-dimensional space and uses a region masking strategy to learn from unlabeled documents. StrucTexTv2 is a document image pre-training framework that performs masked visual text prediction. We use different models to train 200 epochs on Chinese and English financial invoice datasets respectively in the same experimental environment, and the results are shown in Tables 1 and 2.

From the overall view of Tables 1 and 2, SGFNet performs well on the Chinese and English financial invoice datasets with better metrics than other models, including 0.45% and 1.57% higher than StrucTexTv2 on the F1 score metrics in Chinese and English, respectively. Multimodal models such as DocFormer and LayoutXLM obtained good accuracy in invoice information extraction, but without taking into account the semantic correlation between data of the same modality, hence there is still room for further improvement in accuracy relative to SGFNet. RoBERTa is trained using only textual modal data, and the results of the accuracy metrics are significantly different from other multimodal models, illustrating the importance of using multimodal data features for training.

To further analyze the different effects of different modules on the model, the proposed model is ablated to verify the effectiveness of each module. Similarly, we conducted the ablation of various modules of SGFNet on Chinese and English financial invoice datasets respectively, and the results are shown in Tables 3 and 4.

In Tables 3 and 4, SG denotes the semantic graph module, AMS denotes the aggregated multimodal self-attention module, and SCK denotes the loss function combining supervised contrast with smoothed KL divergence. We ablate the above modules separately to observe the experimental results. We ablate the above modules separately to



(a) (b) (c)

Fig. 5. Schematic diagram of preprocessing image. (a) Original image; (b) Cropped original image; (c) Brightened original image.

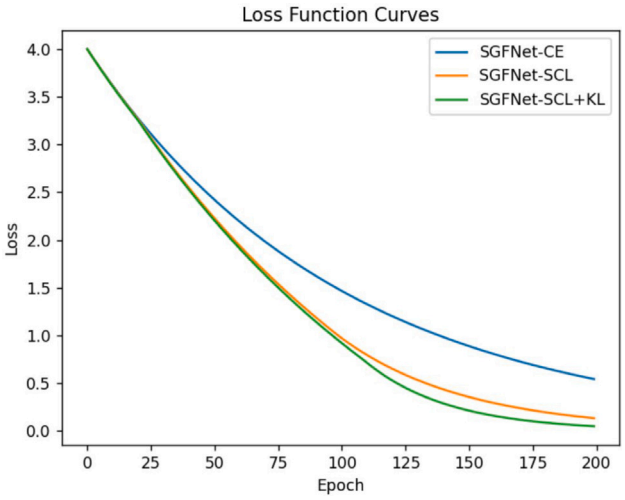


Fig. 6. Comparison chart of loss function curves.

Table 3
Results of ablation experiments on the Chinese invoice dataset.

Model	Precision (%)	Recall (%)	F1 score (%)
SGFNet(+SG+AMS)	94.9	96.17	95.53
SGFNet(+SG+SCK)	94.35	95.31	94.83
SGFNet(+AMS+SCK)	91.21	93.05	92.12
SGFNet(+SG+AMS+SCK)	95.06	97.51	96.27

Table 4
Results of ablation experiments on the English invoice dataset.

Model	Precision (%)	Recall (%)	F1 score (%)
SGFNet(+SG+AMS)	92.02	93.72	92.86
SGFNet(+SG+SCK)	91.43	92.84	92.13
SGFNet(+AMS+SCK)	89.42	91.28	90.34
SGFNet(+SG+AMS+SCK)	93.37	94.05	93.71

observe the experimental results. From the two tables, it can be seen that when a single module is ablated, it leads to a decrease in the metrics. When the loss function is the cross-entropy loss function, the F1 score decreases by 0.74% and 0.85% on the Chinese and English datasets, indicating that the loss function combining supervised contrast and smoothed KL divergence makes the convergence more stable and thus reduces the information loss. When the original self-attention mechanism is employed, the F1 scores decreased by 1.44% and 1.58%, respectively, indicating that the use of the aggregated multimodal self-attention mechanism enhances the feature fusion capability of the network. When the semantic graph module is not employed, the F1 scores drop the most, 4.15% and 3.37%, respectively, indicating that the model guided by the semantic graph can tap into deeper feature representations and thus obtain better information extraction results. Meanwhile, comparing Table 2, we can see that the metrics of SGFNet without the semantic graph module are similar to those of the best-performing baseline model, demonstrating the importance of incorporating semantic information for model performance.

Table 5
Analysis of consumption time of the inference computation.

Model	Parameters (M)	Time (s)
SGFNet	180	0.221
LayoutXLM	200	0.346
StrucTexTv2	238	0.237
LayoutLM	160	0.146

We also analyze the inference computation time for different models and the results are shown in Table 5, where Time denotes the time taken to infer each invoice. From the table, it can be seen that SGFNet obtains a better inference speed with a lower number of parameters compared to LayoutXLM and StrucTexTv2 which have excellent accuracy results. While LayoutLM is faster than SGFNet in inference speed, it has a large gap with SGFNet in accuracy, demonstrating that SGFNet can be efficient in invoice information extraction compared to other baseline models.

5. Conclusion

We propose a semantic graph-based multimodal network, SGFNet, to solve the problem of invoice information extraction in the financial industry. Our data are collected from real financial invoices, which are evaluated and studied after constructing the dataset through preprocessing operations. Firstly, we construct a semantic graph based on text content relevance and guide the fusion and embedding of multimodal data based on the semantic graph. Then the semantically related multimodal information is fed into an aggregated multimodal self-attention module to establish effective connections among the modalities. Finally, a loss function combining supervised contrast learning and smoothed KL divergence is used to achieve information classification. The experimental results on English and Chinese financial invoice datasets show that the F1 score of the proposed method can reach 93.71% and 96.27%, respectively, indicating that SGFNet can extract the information in financial invoices more accurately.

The proposed methodology simultaneously enhances the interaction between intra- and intermodal data, generating richer feature representations, improving the feature extraction capability of the model, and reducing information loss, thereby improving overall feature extraction performance. However, the potential limitations of our study are that the data language is not abundant enough and there is room for further improvement in accuracy. The next step of our research will be to further optimize the network architecture in terms of generalization and accuracy so that the network model can be applied to financial invoice extraction tasks in more language categories and thus better applied to information management decisions in the financial industry.

CRediT authorship contribution statement

Shun Luo: Methodology, Investigation, Validation, Software, Writing – original draft. **Juan Yu:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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